

ADVANCE

# SIMULATED ADVANCE — certification and customer-qualification support for the incumbent GL line

The simulated Ministry-grade evidence resolves an application/qualification gap: the incumbent process can make the target Al-Zn coated coil, but the target standard and customer qualification are not yet certified; a bounded qualification programme removes the binding constraint at minimum support.

SAU-H6-721061 · Certification, testing and customer-qualification support (route 3) · Mode: Ministry simulation

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Aluminium-zinc-alloy plated or coated flat-rolled iron or non-alloy steel, width 600 mm or more  
HS H6 / 721061

Unchanged public conclusion

INVESTIGATE — binding constraint unresolved

|                                                |                                           |
|------------------------------------------------|-------------------------------------------|
| Target-specification demand<br>48 kt           | Effective qualified capacity<br>24.872 kt |
| Published capability distance D*<br>0.25 ratio | Minimum effective support S*<br>5 SAR m   |

Decision conditions

Achieve third-party certification to the target standard and first customer acceptance within 12 months. [1 more; see the full record](#)

Kill conditions

Committed target-specification demand falls below the minimum efficient qualification volume. [1 more; see the full record](#)

Evidence passport ledger: [P-COMTRADE-721061-M](#) [P-COMTRADE-721061-PARTNERS](#)

Snapshot: PUBLIC-SAU-H6-721061-2026-09-12 · As of: 2026-09-12

## Supporting evidence pack

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## Identity and specification boundary

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### Unchanged public conclusion

#### INVESTIGATE — binding constraint unresolved

Route not yet determined

No route hypothesis can be preferred from public evidence; resolve the named route evidence before any support decision.

#### Decision conditions

- No greenfield or financial support recommendation before effective capacity and target specification are resolved.

#### Kill conditions

- Demand is temporary, re-exported or below efficient scale
- Incumbent expansion already closes the specification-adjusted gap

#### Next evidence actions

- Saudi tariff-line and invoice description
- Importer/offtaker application and qualification requirements
- Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages

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- Route feasibility, additionality, policy and downside economics evidence for the candidate routes
- Importer/offtaker application and qualification requirements

## Product identity

| Source field                    | Value                                                                                                                                                                                                                | Unit             |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| HS revision                     | H6                                                                                                                                                                                                                   |                  |
| HS code                         | 721061                                                                                                                                                                                                               |                  |
| National tariff line            | Unavailable<br>Not established by the supplied evidence.                                                                                                                                                             |                  |
| Commercial name, English source | Aluminium-zinc-alloy plated or coated flat-rolled iron or non-alloy steel, width 600 mm or more                                                                                                                      |                  |
| Commercial name, Arabic         | منتجات مسطحة من الحديد أو الصلب غير المخلوط، مطلية بسبائك الألومنيوم والزنك، بعرض 600 مم أو أكثر                                                                                                                     |                  |
| Application boundary            | HS 7210.61 identifies aluminium-zinc-alloy plated or coated flat-rolled iron or non-alloy steel of width 600 mm or more; target coating mass, dimensions, standard, application and qualification remain unresolved. |                  |
| Target specification            | AZ150 aluminium-zinc coated steel coil for roofing and cladding panels<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                                                           |                  |
| Standard                        | SASO-ASTM A792/A792M (synthetic design basis)<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                                                                                    |                  |
| Alloy                           | Unavailable<br>Not established by the supplied evidence.<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                                                                         |                  |
| Temper                          | Unavailable<br>Not established by the supplied evidence.<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                                                                         |                  |
| Application                     | pre-engineered building roofing and cladding<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                                                                                     |                  |
| Zinc coating mass               | 150<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                                                                                                                              | g/m <sup>2</sup> |
| Tin coating mass                | Unavailable<br>Not established by the supplied evidence.<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                                                                         | g/m <sup>2</sup> |

| Source field                    | Value                                                                                                                        | Unit |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------|------|
| Thickness                       | • 0.4• 0.8<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                               | mm   |
| Thickness                       | Unavailable<br>Not established by the supplied evidence.<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a> | µm   |
| Width                           | • 900• 1250<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                              | mm   |
| Customer qualification required | Yes<br><a href="#">SYN-MINISTRY-GALVALUME-001::target_specification</a>                                                      |      |

Classification revision and concordance

Unavailable

Specification source spans

Unavailable

Demand and public trade signals

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Demand conclusion

Gross imports are a screening signal, not demand at the target specification.

| Source field                | Value                                                    | Unit |
|-----------------------------|----------------------------------------------------------|------|
| Base demand                 | 93<br><a href="#">SYN-MINISTRY-GALVALUME-001::demand</a> | kt   |
| Target-specification demand | 48<br><a href="#">SYN-MINISTRY-GALVALUME-001::demand</a> | kt   |
| Downside demand             | 46<br><a href="#">SYN-MINISTRY-GALVALUME-001::demand</a> | kt   |

| Source field           | Value                                                     | Unit                |
|------------------------|-----------------------------------------------------------|---------------------|
| Committed demand       | 30<br><a href="#">SYN-MINISTRY-GALVALUME-001::demand</a>  | kt                  |
| Announced demand       | 12<br><a href="#">SYN-MINISTRY-GALVALUME-001::demand</a>  | kt                  |
| Commitment probability | 0.8<br><a href="#">SYN-MINISTRY-GALVALUME-001::demand</a> | ratio               |
| Observation year       | 2024                                                      | year                |
| Gross import value     | 71.149266                                                 | USD m               |
| Gross import quantity  | 93.232034                                                 | kt                  |
| Named buyers           | Unavailable<br>Not established by the supplied evidence.  |                     |
| Buyer concentration    | Unavailable<br>Not established by the supplied evidence.  |                     |
| Demand timing          | Unavailable<br>Not established by the supplied evidence.  |                     |
| Year                   | Import value, USD m                                       | Import quantity, kt |
| 2022                   | 62.27979                                                  | 53.665168           |
| 2023                   | 90.031058                                                 | 85.840489           |
| 2024                   | 71.149266                                                 | 93.232034           |

## Trade coverage and measurement

|                                              |                                                                                                                                                                                      |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Domestic-origin exports separated            | No                                                                                                                                                                                   |
| Execution ceiling                            | DEGRADED where continuity or tariff-line detail is required; 2021 (H5) excluded by the latest-revision H6 rule Partner detail: PARTNER_DETAIL_OBSERVED (P-COMTRADE-721061-PARTNERS). |
| Trade-flow basis                             | gross                                                                                                                                                                                |
| Missing years                                | <ul style="list-style-type: none"> <li>None recorded; no completed review is implied.</li> </ul>                                                                                     |
| Monthly partner/tariff-line detail available | No                                                                                                                                                                                   |
| Quantity is comparable                       | Yes                                                                                                                                                                                  |

|                      |    |
|----------------------|----|
| Re-exports separated | No |
|----------------------|----|

## Production, retained flows and re-exports

|                              |                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------|
| Domestic-origin exports (kt) | Unavailable                                                                                      |
| Domestic production (kt)     | Unavailable                                                                                      |
| Reference year               | 2024                                                                                             |
| Re-exports (kt)              | Unavailable                                                                                      |
| Retained imports (kt)        | Unavailable                                                                                      |
| Source evidence references   | <ul style="list-style-type: none"> <li>None recorded; no completed review is implied.</li> </ul> |

## Supplier diagnostics

|                                          |                |
|------------------------------------------|----------------|
| Partner concentration by value (HHI)     | 0.7122         |
| Largest supplier share                   | 0.8337         |
| Top-two supplier value share             | 0.9623         |
| Partner concentration by quantity (HHI)  | 0.7471         |
| Largest supplier quantity share          | 0.8566         |
| Bulk unit-value range (USD/t)            | Not calculable |
| Small-volume high-unit-value observation | Not calculable |

## Partner detail (2024 imports)

Observed from un\_comtrade — 7 partner rows

|                                |                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acquisition attempt references | <ul style="list-style-type: none"> <li>P-COMTRADE-721061-PARTNERS-ATTEMPT</li> <li>P-COMTRADE-721061-PARTNERS-ATTEMPT-2</li> <li>P-WITS-721061-PARTNERS-ATTEMPT</li> </ul> |
| Observed partner rows          | 7                                                                                                                                                                          |
| Observed evidence passport     | P-COMTRADE-721061-PARTNERS                                                                                                                                                 |
| Partner snapshot identifier    | PARTNERS-SAU-UN-COMTRADE-2026-09-13                                                                                                                                        |
| Source explanation             | Unavailable                                                                                                                                                                |

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|                   |                         |
|-------------------|-------------------------|
| Source identifier | un_comtrade             |
| State             | PARTNER_DETAIL_OBSERVED |
| Recorded unit     | • 721061• imports• 2024 |

## Domestic supply and qualified capacity

### Supply conclusion

Public producer observations; nameplate is not qualified capacity.

| Named producer | Value |
|----------------|-------|
|----------------|-------|

No named producer evidence is available.

### Observed adjacency signals

- None recorded; no completed review is implied.

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| Source field                 | Value                                                          | Unit  |
|------------------------------|----------------------------------------------------------------|-------|
| Simulated nameplate capacity | 120<br><a href="#">SYN-MINISTRY-GALVALUME-001::plant_line</a>  | kt    |
| Availability factor          | 0.9<br><a href="#">SYN-MINISTRY-GALVALUME-001::plant_line</a>  | ratio |
| Yield factor                 | 0.94<br><a href="#">SYN-MINISTRY-GALVALUME-001::plant_line</a> | ratio |
| Qualification share          | 0.35<br><a href="#">SYN-MINISTRY-GALVALUME-001::plant_line</a> | ratio |
| Market allocation share      | 0.7<br><a href="#">SYN-MINISTRY-GALVALUME-001::plant_line</a>  | ratio |
| Current utilisation          | 0.86<br><a href="#">SYN-MINISTRY-GALVALUME-001::plant_line</a> | ratio |

| Source field                 | Value                                                                                                                                                                  | Unit   |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Description                  | Qualification and certification programme raising the qualified share of the existing GL line; no new nameplate<br><a href="#">SYN-MINISTRY-GALVALUME-001::upgrade</a> |        |
| Incremental capacity         | 28<br><a href="#">SYN-MINISTRY-GALVALUME-001::upgrade</a>                                                                                                              | kt     |
| Upgrade schedule             | 12<br><a href="#">SYN-MINISTRY-GALVALUME-001::upgrade</a>                                                                                                              | months |
| Effective qualified capacity | 24.872                                                                                                                                                                 | kt     |
| Qualified available supply   | 24.8724                                                                                                                                                                | kt     |

## Gap diagnosis and false-positive controls

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Application gap

| Source field               | Value  | Unit |
|----------------------------|--------|------|
| Specification-adjusted gap | 23.128 | kt   |

| Source reference             | Status        | Value                                                                                                                                                                                                 |
|------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EX-01_HETEROGENEOUS_RESIDUAL | Not satisfied | The commercial product cannot be separated from a highly heterogeneous residual code and no product-level evidence is available.<br><a href="#">SYN-MINISTRY-GALVALUME-001::hard exclusion inputs</a> |
| EX-02_MARKET_BELOW_MES       | Not satisfied | The sustainable downside market is below minimum efficient scale and no credible export contract exists.<br><a href="#">SYN-MINISTRY-GALVALUME-001::hard exclusion inputs</a>                         |

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| Source reference                | Status        | Value                                                                                                                                                                                               |
|---------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EX-03_UNSATISFIABLE_HARD_GATE   | Not satisfied | A legal, safety, environmental, IP or customer-qualification hard gate cannot be satisfied.<br><a href="#">SYN-MINISTRY-GALVOLUME-001::hard_exclusion_inputs</a>                                    |
| EX-04_IDLE_EQUIVALENT_CAPACITY  | Not satisfied | Domestic production meets the required specification with material idle capacity and no binding market failure is evidenced.<br><a href="#">SYN-MINISTRY-GALVOLUME-001::hard_exclusion_inputs</a>   |
| EX-05_TRANSITORY_OR_MEASUREMENT | Not satisfied | The apparent gap is dominated by re-export, one-off project demand, temporary price arbitrage or classification discontinuity.<br><a href="#">SYN-MINISTRY-GALVOLUME-001::hard_exclusion_inputs</a> |
| EX-06_REDUNDANCY_OR_CROWD_OUT   | Not satisfied | The intervention would create unacceptable redundant capacity or crowd out a more efficient incumbent.<br><a href="#">SYN-MINISTRY-GALVOLUME-001::hard_exclusion_inputs</a>                         |

### Test the apparent gap

| Source reference         | Status        | Value                                                                                                     |
|--------------------------|---------------|-----------------------------------------------------------------------------------------------------------|
| hard_exclusion           | Not satisfied | HARD_EXCLUSION_NOT_SATISFIED<br><a href="#">SYN-MINISTRY-GALVOLUME-001::hard_exclusion_inputs</a>         |
| false_or_measurement_gap | Not satisfied | FALSE_OR_MEASUREMENT_NOT_ESTABLISHED<br><a href="#">SYN-MINISTRY-GALVOLUME-001::hard_exclusion_inputs</a> |

| Source reference                 | Status        | Value                                                                                                                                                        |
|----------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| equivalent_idle_qualified_supply | Not satisfied | EQUIVALENT_QUALIFIED_SUPPLY_NOT_ESTABLISHED<br><a href="#">SYN-MINISTRY-GALVOLUME-001::demand</a><br><a href="#">SYN-MINISTRY-GALVOLUME-001::equivalence</a> |
| uneconomic_at_efficient_scale    | Not satisfied | ECONOMIC_ROUTE_EXISTS<br><a href="#">SYN-MINISTRY-GALVOLUME-001::route_evidence</a>                                                                          |
| structural_overcapacity          | Not satisfied | STRUCTURAL_OVERCAPACITY_NOT_ESTABLISHED<br><a href="#">SYN-MINISTRY-GALVOLUME-001::hard_exclusion_inputs</a>                                                 |
| generic_capacity_contradicted    | Not satisfied | GENERIC_CAPACITY_NOT_CONTRADICTED<br><a href="#">P-COMTRADE-721061-X</a>                                                                                     |

## Capability and route alternatives

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| Source field                     | Value | Unit  |
|----------------------------------|-------|-------|
| Known coverage K                 | 1     | ratio |
| Unknown weight U                 | 0     | ratio |
| Uncertainty penalty (lambda)     | 0.5   | ratio |
| Minimum known coverage           | 0.7   | ratio |
| Published capability distance D* | 0.25  | ratio |

| Source field        | State | Weight |
|---------------------|-------|--------|
| Feedstock chemistry | 0     | 0.05   |
| Core process route  | 0     | 0.2    |

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| Source field                             | State                                                                                            | Weight |
|------------------------------------------|--------------------------------------------------------------------------------------------------|--------|
| Equipment operating envelope             | 1                                                                                                | 0.2    |
| Finishing and specification control      | 1                                                                                                | 0.15   |
| Quality, laboratories and metrology      | 1                                                                                                | 0.1    |
| Certification and customer qualification | 2                                                                                                | 0.1    |
| Capacity and timing                      | 1                                                                                                | 0.1    |
| Utilities, environment and permitting    | 0                                                                                                | 0.05   |
| Skills and market integration            | 0                                                                                                | 0.05   |
| Sector-profile hard gates                |                                                                                                  |        |
| Substrate range                          | RESOLVED                                                                                         |        |
| Width/thickness envelope                 | RESOLVED                                                                                         |        |
| Coating process and mass                 | RESOLVED                                                                                         |        |
| Surface treatment                        | RESOLVED                                                                                         |        |
| Mandatory or customer standard           | RESOLVED                                                                                         |        |
| Unresolved hard gates                    | <ul style="list-style-type: none"> <li>None recorded; no completed review is implied.</li> </ul> |        |
| Known hard-gate failures                 | <ul style="list-style-type: none"> <li>None recorded; no completed review is implied.</li> </ul> |        |

## No intervention

Fails

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Passes         |
| Resolves the binding constraint | Fails          |
| Additionality                   | Passes         |
| Policy permissibility           | Passes         |
| Unsupported NPV                 | Not calculable |
| Unsupported IRR                 | Not calculable |
| Minimum effective support S*    | Not calculable |

**Lower-route precedence:** No lower route recorded

**Incremental national value** Not calculable

passes

- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

## Administrative or regulatory correction

Fails

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |
| Unsupported NPV                 | Not calculable |
| Unsupported IRR                 | Not calculable |
| Minimum effective support S*    | Not calculable |

**Lower-route precedence:** No lower route recorded

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**Incremental national value** Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

## Information and market linkage

Fails

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |

**Unsupported NPV** Not calculable

**Unsupported IRR** Not calculable

**Minimum effective support S\*** Not calculable

**Lower-route precedence:** No lower route recorded

**Incremental national value** Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.

## Certification, testing and quality

Passes

| Source field                    | Value  |
|---------------------------------|--------|
| Feasibility                     | Passes |
| Resolves the binding constraint | Passes |
| Additionality                   | Passes |
| Policy permissibility           | Passes |

**Unsupported NPV** -4.743

|                                                        |         |
|--------------------------------------------------------|---------|
| Unsupported IRR                                        | 0.08686 |
| Minimum effective support S*                           | 5       |
| <b>Lower-route precedence:</b> No lower route recorded |         |
| Incremental national value                             | 82      |
| Capacity ratio                                         | 1.1494  |
| Warning threshold                                      | 1.25    |
| Passes the recorded test                               | Yes     |

[SYN-MINISTRY-GALVALUME-001::route\\_evidence](#)

## Demand aggregation and offtake

Fails

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |
| Unsupported NPV                 | Not calculable |
| Unsupported IRR                 | Not calculable |
| Minimum effective support S*    | Not calculable |

**Lower-route precedence:** Certification, testing and quality

|                            |                |
|----------------------------|----------------|
| Incremental national value | Not calculable |
|----------------------------|----------------|

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

## Brownfield upgrade or expansion

Fails

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |
| Unsupported NPV                 | Not calculable |
| Unsupported IRR                 | Not calculable |
| Minimum effective support S*    | Not calculable |

**Lower-route precedence:** Certification, testing and quality

**Incremental national value** Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

## Technology partnership or joint venture

Fails

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |
| Unsupported NPV                 | Not calculable |
| Unsupported IRR                 | Not calculable |
| Minimum effective support S*    | Not calculable |

**Lower-route precedence:** Certification, testing and quality

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**Incremental national value**      Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

## Greenfield capacity

Fails

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |

**Unsupported NPV**      Not calculable

**Unsupported IRR**      Not calculable

**Minimum effective support S\***      Not calculable

**Lower-route precedence:** Certification, testing and quality

**Incremental national value**      Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

## Shared enabling infrastructure

Not calculable

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |

|                              |                |
|------------------------------|----------------|
| Unsupported NPV              | Not calculable |
| Unsupported IRR              | Not calculable |
| Minimum effective support S* | Not calculable |

**Lower-route precedence:** No lower route recorded

|                            |                |
|----------------------------|----------------|
| Incremental national value | Not calculable |
|----------------------------|----------------|

Not calculable

- GRAPH\_REQUIRED: route 8 requires the governed dependency graph and a positive UnlockValue.

Public route comparison

No intervention

Not calculable

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |

|                              |                |
|------------------------------|----------------|
| Unsupported NPV              | Not calculable |
| Unsupported IRR              | Not calculable |
| Minimum effective support S* | Not calculable |

**Lower-route precedence:** No lower route recorded

|                            |                |
|----------------------------|----------------|
| Incremental national value | Not calculable |
|----------------------------|----------------|

Not calculable

- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

## Administrative or regulatory correction

Not calculable

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |
| Unsupported NPV                 | Not calculable |
| Unsupported IRR                 | Not calculable |
| Minimum effective support S*    | Not calculable |

**Lower-route precedence:** No lower route recorded

**Incremental national value** Not calculable

Not calculable

- Administrative, classification or regulatory barrier evidence is required.

## Information and market linkage

Not calculable

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |
| Unsupported NPV                 | Not calculable |
| Unsupported IRR                 | Not calculable |
| Minimum effective support S*    | Not calculable |

**Lower-route precedence:** No lower route recorded

**Incremental national value** Not calculable

Not calculable

- A verified information, linkage or matching constraint is required.

## Certification, testing and quality

Not calculable

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |

**Unsupported NPV** Not calculable

**Unsupported IRR** Not calculable

**Minimum effective support S\*** Not calculable

**Lower-route precedence:** No lower route recorded

**Incremental national value** Not calculable

Not calculable

- A verified specification, certification, testing or qualification constraint is required.

## Demand aggregation and offtake

Not calculable

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |

**Unsupported NPV** Not calculable

|                                                                                                                      |                |
|----------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Unsupported IRR</b>                                                                                               | Not calculable |
| <b>Minimum effective support S*</b>                                                                                  | Not calculable |
| <b>Lower-route precedence:</b> No lower route recorded                                                               |                |
| <b>Incremental national value</b>                                                                                    | Not calculable |
| Not calculable                                                                                                       |                |
| <ul style="list-style-type: none"> <li>A verified demand-fragmentation or offtake constraint is required.</li> </ul> |                |

## Brownfield upgrade or expansion

| Not calculable                                                                                                                                                             |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Source field                                                                                                                                                               | Value          |
| <b>Feasibility</b>                                                                                                                                                         | Not calculable |
| <b>Resolves the binding constraint</b>                                                                                                                                     | Not calculable |
| <b>Additionality</b>                                                                                                                                                       | Not calculable |
| <b>Policy permissibility</b>                                                                                                                                               | Not calculable |
| <b>Unsupported NPV</b>                                                                                                                                                     | Not calculable |
| <b>Unsupported IRR</b>                                                                                                                                                     | Not calculable |
| <b>Minimum effective support S*</b>                                                                                                                                        | Not calculable |
| <b>Lower-route precedence:</b> No lower route recorded                                                                                                                     |                |
| <b>Incremental national value</b>                                                                                                                                          | Not calculable |
| Not calculable                                                                                                                                                             |                |
| <ul style="list-style-type: none"> <li>Verified incumbent adjacency makes brownfield the priority hypothesis; effective capacity and economics remain required.</li> </ul> |                |

## Technology partnership or joint venture

Not calculable

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S\* Not calculable

**Lower-route precedence:** No lower route recorded

Incremental national value Not calculable

Not calculable

- A verified technology constraint and major-line/JV adjacency are required.

## Greenfield capacity

Not calculable

| Source field                    | Value          |
|---------------------------------|----------------|
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S\* Not calculable

**Lower-route precedence:** No lower route recorded

|                                                                                                                                                                        |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Incremental national value                                                                                                                                             | Not calculable |
| Not calculable                                                                                                                                                         |                |
| <ul style="list-style-type: none"> <li>Greenfield requires failed lower routes, D* above the configured band, demand at MES, and passing competition gates.</li> </ul> |                |

Shared enabling infrastructure

| Not calculable                  |                |
|---------------------------------|----------------|
| Source field                    | Value          |
| Feasibility                     | Not calculable |
| Resolves the binding constraint | Not calculable |
| Additionality                   | Not calculable |
| Policy permissibility           | Not calculable |
| Unsupported NPV                 | Not calculable |
| Unsupported IRR                 | Not calculable |
| Minimum effective support S*    | Not calculable |

Lower-route precedence: No lower route recorded

|                                                                                                                                              |                |
|----------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Incremental national value                                                                                                                   | Not calculable |
| Not calculable                                                                                                                               |                |
| <ul style="list-style-type: none"> <li>GRAPH_REQUIRED: route 8 requires the governed dependency graph and a positive UnlockValue.</li> </ul> |                |

Counterfactual

|                                                                      |     |
|----------------------------------------------------------------------|-----|
| Can the incumbent meet the specification without capital investment? | No  |
| Does downside demand support both incumbent and entrant?             | No  |
| Does entry displace efficient domestic production?                   | Yes |

|                                                                  |                                                                                                                                                                                                      |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is a technology partnership superior to expansion or greenfield? | No                                                                                                                                                                                                   |
| Missing capabilities                                             | <ul style="list-style-type: none"><li>equipment_envelope</li><li>finishing_spec_control</li><li>qa_lab_metrology</li><li>certification_customer_qualification</li><li>capacity_time_window</li></ul> |
| Brownfield versus greenfield comparison                          |                                                                                                                                                                                                      |
| Incremental capacity                                             | 28                                                                                                                                                                                                   |
| Upgrade schedule                                                 | 12                                                                                                                                                                                                   |
| Unsupported NPV                                                  | -4.743                                                                                                                                                                                               |
| Minimum effective support S*                                     | 5                                                                                                                                                                                                    |
| Greenfield alternative                                           | Unavailable                                                                                                                                                                                          |

## Economics and intervention

Class D simulation · SYN-MINISTRY-GALVALUME-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

| Source field                 | Value                                                              | Unit  |
|------------------------------|--------------------------------------------------------------------|-------|
| Unsupported NPV              | -4.743                                                             | SAR m |
| Unsupported IRR              | 0.08686                                                            | ratio |
| Minimum effective support S* | 5                                                                  | SAR m |
| Supported NPV                | 0.257                                                              | SAR m |
| Supported IRR                | 0.12199                                                            | ratio |
| Hurdle rate                  | 0.12                                                               | ratio |
| Separate downside result     | Not calculable<br>Cannot be calculated from the supplied evidence. | SAR m |

### National-value comparison

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة  
SIMULATED — NOT MINISTRY EVIDENCE

|                            |     |
|----------------------------|-----|
| National benefits          | 108 |
| National costs             | 26  |
| Incremental national value | 82  |
| Positive result            | Yes |
| Components                 |     |
| Displacement               | 8   |
| Domestic value added       | 64  |
| Exports                    | 6   |
| Fiscal receipts            | 11  |
| Government cost            | 5   |
| Knowledge and skills       | 9   |
| Resilience value           | 18  |
| Resources and environment  | 3   |
| Risk allowance             | 10  |

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Cash flows without support (SAR m), in source order • -42• 7.5• 7.5• 7.5• 7.5• 7.5• 7.5• 7.5• 7.5

Currency SAR\_m

Hurdle rate 0.12

Minimum efficient scale (kt) 30

National value

Displacement 8

Domestic value added 64

Exports 6

Fiscal receipts 11

Government cost 5

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة  
SIMULATED — NOT MINISTRY EVIDENCE

|                           |    |
|---------------------------|----|
| Knowledge and skills      | 9  |
| Resilience value          | 18 |
| Resources and environment | 3  |
| Risk allowance            | 10 |

|                    |                                                              |
|--------------------|--------------------------------------------------------------|
| Support instrument | certification, testing and customer-qualification cost-share |
|--------------------|--------------------------------------------------------------|

[SYN-MINISTRY-GALVALUME-001::economics](#)

**Cash-flow inputs in source order**

|                                              |                                                                                                                                                                          |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Route code                                   | 3                                                                                                                                                                        |
| Downside cash flows (SAR m), in source order | <div> <div>◦</div> <div>-42◦</div> <div>7.5◦</div> <div>7.5◦</div> <div>7.5◦</div> <div>7.5◦</div> <div>7.5◦</div> <div>7.5◦</div> <div>7.5◦</div> <div>7.5</div> </div> |
| Hurdle rate                                  | 0.12                                                                                                                                                                     |
| Recorded basis                               | Certification, testing and customer-qualification programme on the incumbent GL line.                                                                                    |

[SYN-MINISTRY-GALVALUME-001::route\\_evidence](#)

**Competition and public policy**

Class D simulation · SYN-MINISTRY-GALVALUME-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

| Source field                      | Value                                                    | Unit  |
|-----------------------------------|----------------------------------------------------------|-------|
| Capacity to downside demand ratio | 1.1494                                                   | ratio |
| Displacement cost                 | 8                                                        | SAR m |
| Sunset date                       | Unavailable<br>Not established by the supplied evidence. |       |
| Capacity to downside demand ratio | 1.1494                                                   |       |

|                                |                            |
|--------------------------------|----------------------------|
| Warning threshold              | 1.25                       |
| Warning is triggered           | No                         |
| Triggers the default warning   | Yes                        |
| Displacement cost              | 8                          |
| Non-additionality assessment   |                            |
| Status                         | passes                     |
| Recorded basis                 | route_record_additionality |
| Concentration before and after |                            |
| Not calculable                 |                            |

Calculation and recommendation do not authorize public support.

## Complete rule execution ledger

### R0 · Identity gate

Full · Fires

HS revision H6 and HS6 721061 are frozen in the snapshot; specification/application remains separately gated.

Continue to screening while preserving the unresolved decision object.

### R1-F · Persistent retained exposure — full

Full · Does not fire

Full persistence test did not fire.

Do not claim the full persistence rule.

**Missing years** • None recorded; no completed review is implied.

### R1-D · Persistence signal — degraded

Degraded · Fires

3 positive observed years within the configured window; no interpolation used.

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة  
SIMULATED — NOT MINISTRY EVIDENCE

Generate INVESTIGATE only; confidence capped at C.

|                         |                    |
|-------------------------|--------------------|
| Positive observed years | • 2022• 2023• 2024 |
| Confidence ceiling      | C                  |

R2 · Quantity-led expansion

Full · Does not fire

Value movement is not quantity-led under the configured test.

Separate structural volume growth from price movement.

|                                      |         |
|--------------------------------------|---------|
| First year                           | 2023    |
| Final year                           | 2024    |
| Observed period (years)              | 1       |
| Change in log value                  | -0.2354 |
| Change in log quantity               | 0.0826  |
| Change in log unit value             | -0.318  |
| Quantity contribution share          | 0.2062  |
| Quantity compound annual growth rate | 0.0861  |

R3 · Supplier concentration

Full · Fires

External supply is concentrated on at least one calculable basis.

Generate a resilience/diversification review, not an automatic localisation recommendation.

|                               |        |
|-------------------------------|--------|
| Concentration index (HHI)     | 0.7122 |
| Largest supplier share        | 0.8337 |
| Top-two supplier share        | 0.9623 |
| Concentration threshold (HHI) | 0.25   |
| Largest-supplier threshold    | 0.5    |

| Value                      |                      |
|----------------------------|----------------------|
| Recorded basis             | value                |
| Status                     | FULL                 |
| Observation year           | 2024                 |
| Trade-flow basis           | gross                |
| Source                     | partner_observations |
| Concentration index (HHI)  | 0.7122               |
| Largest supplier share     | 0.8337               |
| Top-two supplier share     | 0.9623               |
| Valid observation coverage | 1                    |
| Eligible partner count     | 7                    |
| Source explanation         | Unavailable          |
| Quantity                   |                      |
| Recorded basis             | quantity             |
| Status                     | FULL                 |
| Observation year           | 2024                 |
| Trade-flow basis           | gross                |
| Source                     | partner_observations |
| Concentration index (HHI)  | 0.7471               |
| Largest supplier share     | 0.8566               |
| Top-two supplier share     | 0.9706               |
| Valid observation coverage | 1                    |
| Eligible partner count     | 7                    |
| Source explanation         | Unavailable          |

---

## R4-F · Product-tier heterogeneity — full

Disabled · Not evaluable

Partner-month × tariff-line cells are unavailable.

Do not fit clusters or claim grades.

---

## R4-D · Unit-value dispersion — degraded

Disabled · Not evaluable

Comparable annual partner coverage is insufficient; R4-D is not calculable.

Open specification research only; no grade conclusion.

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| Status                          | Not calculable                                                    |
| Calculation basis               | partner_observations                                              |
| Valid value coverage            | 1                                                                 |
| Valid quantity coverage         | 1                                                                 |
| Minimum valid-value coverage    | 0.7                                                               |
| Weighted first quartile (USD/t) | Not calculable                                                    |
| Weighted median (USD/t)         | Not calculable                                                    |
| Weighted third quartile (USD/t) | Not calculable                                                    |
| Interquartile range (USD/t)     | Not calculable                                                    |
| Bulk unit-value band (USD/t)    | Not calculable                                                    |
| Potential outlier               | Not calculable                                                    |
| Source explanation              | Comparable annual partner coverage or dispersion is insufficient. |

---

## R5 · Domestic supply plus continued imports

Disabled · Not evaluable

Domestic capability or a usable import measure is unavailable.

Test specification, qualification, capacity, price, application and allocation mismatch.

|                       |                |
|-----------------------|----------------|
| Retained imports (kt) | Not calculable |
|-----------------------|----------------|

---

|                                                |                                                                                                                                                        |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Net import exposure (kt)</b>                | Not calculable                                                                                                                                         |
| <b>Apparent consumption (kt)</b>               | Not calculable                                                                                                                                         |
| <b>Retained imports / apparent consumption</b> | Not calculable                                                                                                                                         |
| <b>Calculation basis</b>                       | Not calculable                                                                                                                                         |
| <b>Unavailable inputs</b>                      | <ul style="list-style-type: none"> <li>domestic_origin_exports_kt • domestic_production_kt</li> <li>reexports_kt • retained_imports_kt</li> </ul>      |
| <b>Reasons for unavailability</b>              |                                                                                                                                                        |
| <b>Retained imports (kt)</b>                   | <ul style="list-style-type: none"> <li>reexports_kt • retained_imports_kt</li> </ul>                                                                   |
| <b>Net import exposure (kt)</b>                | <ul style="list-style-type: none"> <li>domestic_origin_exports_kt • retained_imports_kt</li> </ul>                                                     |
| <b>Apparent consumption (kt)</b>               | <ul style="list-style-type: none"> <li>domestic_origin_exports_kt • domestic_production_kt</li> <li>retained_imports_kt</li> </ul>                     |
| <b>Retained imports / apparent consumption</b> | <ul style="list-style-type: none"> <li>apparent_consumption_kt • retained_imports_kt</li> </ul>                                                        |
| <b>Configured threshold</b>                    | 0.2                                                                                                                                                    |
| <b>Source explanation</b>                      | Domestic production quantity and retained-import flow are absent from the frozen public snapshot; gross imports cannot establish apparent consumption. |

## R6 · Capacity pressure

Disabled · Not evaluable

Effective utilisation and target-specification shortage are not public.

Obtain line-level availability, yield, qualification share and allocation.

## R7 · Latent domestic capacity

Disabled · Not evaluable

Idle qualified capacity and specification equivalence are not established publicly.

Do not assume latent capacity.

---

## R8 · Committed future demand

Disabled · Not evaluable

Awarded/financed demand at the target specification is unavailable.

Keep base, committed and announced demand separate when provided.

---

## R9-S · Coarse incumbent adjacency screen

Disabled · Not evaluable

No defensible coarse adjacency signal.

Open the full line-level capability assessment; do not publish D\* from screening alone.

|                     |                |
|---------------------|----------------|
| Same process family | Not calculable |
|---------------------|----------------|

---

|                         |   |
|-------------------------|---|
| Qualifying signal count | 0 |
|-------------------------|---|

---

|                         |    |
|-------------------------|----|
| Known hard-gate failure | No |
|-------------------------|----|

---

## R10 · Strategic criticality

Degraded · Fires

A resilience review is warranted by concentration; formal criticality still requires authority confirmation.

Keep commercial viability and strategic value separate.

|                    |                |
|--------------------|----------------|
| Criticality status | Not calculable |
|--------------------|----------------|

---

|                                |             |
|--------------------------------|-------------|
| Criticality evidence reference | Unavailable |
|--------------------------------|-------------|

---

## R11 · Economic exclusion / generic-capacity warning

Degraded · Does not fire

Export/import ratio or established nameplate capability is NOT\_CALCULABLE; no generic-capacity exclusion fires.

Reject generic support or move only a named specialty exception to investigation.

|                           |        |
|---------------------------|--------|
| Export/import value ratio | 0.0002 |
|---------------------------|--------|

---

|                               |    |
|-------------------------------|----|
| Export/import ratio threshold | 50 |
|-------------------------------|----|

---

|                                          |                                                                                                    |
|------------------------------------------|----------------------------------------------------------------------------------------------------|
| Calculated export/import value ratio     | 0.0002                                                                                             |
| Disclosed export/import value ratio      | Unavailable                                                                                        |
| Disclosed ratio is consistent            | Unavailable                                                                                        |
| Ratio basis                              | gross_trade_value                                                                                  |
| Ratio status                             | CALCULABLE                                                                                         |
| Ratio explanation                        | Unavailable                                                                                        |
| Established nominal capacity             |                                                                                                    |
| Established by evidence                  | No                                                                                                 |
| Total observed nominal capacity (t/year) | Not calculable                                                                                     |
| Named producers                          | <ul style="list-style-type: none"> <li>• None recorded; no completed review is implied.</li> </ul> |
| Evidence references                      | <ul style="list-style-type: none"> <li>• None recorded; no completed review is implied.</li> </ul> |
| Source explanation                       | Verified domestic capability is not established.                                                   |

## R12 · Evidence-value trigger

Degraded · Fires

5 named facts could change the route.

Prioritise the smallest evidence request with a plausible route effect; quantify EVSI when cost and value inputs exist.

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Named missing facts | <ul style="list-style-type: none"> <li>• Saudi tariff-line and invoice description</li> <li>• Importer/offtaker application and qualification requirements</li> <li>• Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages</li> <li>• Route feasibility, additionality, policy and downside economics evidence for the candidate routes</li> <li>• Importer/offtaker application and qualification requirements</li> </ul> |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Evidence needed

- Saudi tariff-line and invoice description

|                                                                                                            |                                                                    |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Evidence-need identifier</b>                                                                            | identity/tariff-line                                               |
| <b>Blocked decision field</b>                                                                              | product_identity                                                   |
| <b>Effect on route</b>                                                                                     | Can resolve the decision object and change the admissible route.   |
| <b>Evidence references</b>                                                                                 | ◦ P-WCO-721061                                                     |
| <b>Numerical value of information</b>                                                                      | Not calculable                                                     |
| • Importer/offtaker application and qualification requirements                                             |                                                                    |
| <b>Evidence-need identifier</b>                                                                            | target specification/application                                   |
| <b>Blocked decision field</b>                                                                              | demand_at_required_specification                                   |
| <b>Effect on route</b>                                                                                     | Can establish target demand and change state or route.             |
| <b>Evidence references</b>                                                                                 | ◦ None recorded; no completed review is implied.                   |
| <b>Numerical value of information</b>                                                                      | Not calculable                                                     |
| • Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages |                                                                    |
| <b>Evidence-need identifier</b>                                                                            | capacity/availability/allocation                                   |
| <b>Blocked decision field</b>                                                                              | domestic_supply_or_capability                                      |
| <b>Effect on route</b>                                                                                     | Can distinguish no action, brownfield expansion, and new capacity. |
| <b>Evidence references</b>                                                                                 | ◦ None recorded; no completed review is implied.                   |
| <b>Numerical value of information</b>                                                                      | Not calculable                                                     |
| • Route feasibility, additionality, policy and downside economics evidence for the candidate routes        |                                                                    |
| <b>Evidence-need identifier</b>                                                                            | route economics                                                    |
| <b>Blocked decision field</b>                                                                              | route_economics                                                    |
| <b>Effect on route</b>                                                                                     | Can determine whether the plausible route proceeds unsupported.    |
| <b>Evidence references</b>                                                                                 | ◦ None recorded; no completed review is implied.                   |
| <b>Numerical value of information</b>                                                                      | Not calculable                                                     |

- Importer/offtaker application and qualification requirements

|                                       |                                                                                                    |
|---------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Evidence-need identifier</b>       | line-level production or producer-grade matrix                                                     |
| <b>Blocked decision field</b>         | hard_regulatory_or_process_gate                                                                    |
| <b>Effect on route</b>                | Can establish domestic equivalence and change the incumbent route.                                 |
| <b>Evidence references</b>            | <ul style="list-style-type: none"> <li>◦ None recorded; no completed review is implied.</li> </ul> |
| <b>Numerical value of information</b> | Not calculable                                                                                     |

## Simulated R6–R8 ledger

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

### R5 · Retained import penetration

Disabled · Not evaluable

Synthetic retained-flow inputs are absent; R5 is not calculable.

Use only when production and retained-flow blocks are declared.

|                                            |                |
|--------------------------------------------|----------------|
| <b>Retained imports (kt)</b>               | Not calculable |
| <b>Domestic production (kt)</b>            | Not calculable |
| <b>Gross import quantity</b>               | 93.232034      |
| <b>Retained-import penetration</b>         | Not calculable |
| <b>Apparent consumption (kt)</b>           | Not calculable |
| <b>Minimum retained-import penetration</b> | 0.2            |

### R6 · Capacity pressure

Degraded · Fires

Synthetic utilisation and shortage proxy meet the configured R6 thresholds; sustained period is NOT\_CALCULABLE.

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة  
SIMULATED — NOT MINISTRY EVIDENCE

Use as a DEGRADED capacity-pressure signal only; it cannot by itself support ADVANCE without a sustained period.

|                                        |                                 |
|----------------------------------------|---------------------------------|
| Effective utilisation                  | 0.86                            |
| Minimum effective utilisation          | 0.85                            |
| Target-specification demand            | 48                              |
| Effective qualified capacity           | 24.872                          |
| Effective-capacity source field        | effective_qualified_capacity_kt |
| Shortage ratio                         | 0.9299                          |
| Shortage denominator                   | effective_qualified_capacity_kt |
| Minimum specification-matched shortage | 0.1                             |
| Sustained period                       | Not calculable                  |

## R7 · Latent domestic capacity

Full · Does not fire

Synthetic utilisation/equivalence does not meet the configured latent-capacity test.

Test no-support, market linkage, procurement, or barrier removal only when equivalence and latent capacity are established.

|                               |                |
|-------------------------------|----------------|
| Effective utilisation         | 0.86           |
| Maximum effective utilisation | 0.7            |
| Specification equivalence     | No             |
| Qualified available supply    | Not calculable |

## R8 · Committed future demand

Full · Fires

Committed and announced layers are disclosed, but base demand, commitment probability, and minimum efficient scale govern R8 calculability.

Do not infer probability-adjusted demand addition or MES fill; keep committed and announced demand separate.

|                                              |          |
|----------------------------------------------|----------|
| Base demand                                  | 93       |
| Committed demand                             | 30       |
| Announced demand                             | 12       |
| Target-specification demand                  | 48       |
| Downside demand                              | 46       |
| Commitment probability                       | 0.8      |
| Probability-adjusted committed demand (kt)   | 24       |
| Probability-adjusted demand addition         | 0.258065 |
| Minimum probability-adjusted demand addition | 0.2      |
| Minimum efficient scale (kt)                 | 30       |
| Minimum efficient scale coverage             | 0.8      |
| Minimum scale-coverage threshold             | 0.25     |

## Evidence passports and contradictions

7 public records; 14 synthetic records.

### Contradiction register

#### Public evidence contradictions

No public contradictions recorded.

#### Synthetic evidence contradictions

No synthetic contradictions are recorded for this simulation.

P-COMTRADE-721061-M

UN Comtrade Saudi imports rows for HS 721061

| Source field           | Value                                                                                                                                                                                                                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source                 | UN Comtrade                                                                                                                                                                                                                                                                                                                                       |
| Class                  | B                                                                                                                                                                                                                                                                                                                                                 |
| Status                 | calculated                                                                                                                                                                                                                                                                                                                                        |
| Source period          | 2022/2023/2024                                                                                                                                                                                                                                                                                                                                    |
| Retrieval date         | 2026-09-12                                                                                                                                                                                                                                                                                                                                        |
| Reviewer status        | unconfirmed_by_responsible_authority                                                                                                                                                                                                                                                                                                              |
| Transformation         | LATEST_REVISION_ONLY H6 rows; trade value divided by 1,000,000 to USD million and net weight divided by 1,000,000 to kt.                                                                                                                                                                                                                          |
| Supporting claims      | <ul style="list-style-type: none"><li>TRADE_QUANTITY</li><li>TRADE_VALUE</li></ul>                                                                                                                                                                                                                                                                |
| Contradiction register | Unavailable                                                                                                                                                                                                                                                                                                                                       |
| Source reference       | <a href="https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=M&amp;cmdCode=AG6&amp;partnerCode=0&amp;motCode=0&amp;customsCode=C00">https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=M&amp;cmdCode=AG6&amp;partnerCode=0&amp;motCode=0&amp;customsCode=C00</a> |

P-COMTRADE-721061-PARTNERS

UN Comtrade Saudi 2024 import partner rows for HS 721061

| Source field      | Value                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source            | UN Comtrade                                                                                                                                                                                 |
| Class             | B                                                                                                                                                                                           |
| Status            | calculated                                                                                                                                                                                  |
| Source period     | 2024                                                                                                                                                                                        |
| Retrieval date    | 2026-09-13                                                                                                                                                                                  |
| Reviewer status   | unconfirmed_by_responsible_authority                                                                                                                                                        |
| Transformation    | World aggregate excluded; value divided by 1,000,000 to USD million and net weight divided by 1,000,000 to kt. Partner detail state: PARTNER_DETAIL_OBSERVED (un_comtrade; 7 partner rows). |
| Supporting claims | <ul style="list-style-type: none"><li>SUPPLIER_CONCENTRATION</li></ul>                                                                                                                      |

| Source field           | Value                                                                                                                                                                                                                                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contradiction register | Unavailable                                                                                                                                                                                                                                                                                                                                                   |
| Source reference       | <a href="https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=M&amp;cmdCode=721061&amp;motCode=0&amp;customsCode=C00&amp;includeDesc=true">https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=M&amp;cmdCode=721061&amp;motCode=0&amp;customsCode=C00&amp;includeDesc=true</a> |

### P-COMTRADE-721061-PARTNERS-ATTEMPT

UN Comtrade Saudi 2024 import partner attempt for HS 721061

| Source field           | Value                                                                                                                                                                                                                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source                 | UN Comtrade                                                                                                                                                                                                                                                                                                         |
| Class                  | B                                                                                                                                                                                                                                                                                                                   |
| Status                 | unresolved                                                                                                                                                                                                                                                                                                          |
| Source period          | 2024                                                                                                                                                                                                                                                                                                                |
| Retrieval date         | 2026-09-13                                                                                                                                                                                                                                                                                                          |
| Reviewer status        | unconfirmed_by_responsible_authority                                                                                                                                                                                                                                                                                |
| Transformation         | PARTNER_DETAIL_ATTEMPT_SUPERSEDED:VARIANT_NOT_TRANSMITTED; unit un_comtrade/e71395bbc925/20260913T014942Z; http_status 200; normalization_status NORMALIZED; superseded by COMPLETE unit un_comtrade/050a03afdc48/20260913T015620Z. This passport evidences the attempt, not trade.                                 |
| Supporting claims      | <ul style="list-style-type: none"> <li>SUPPLIER_CONCENTRATION</li> </ul>                                                                                                                                                                                                                                            |
| Contradiction register | Unavailable                                                                                                                                                                                                                                                                                                         |
| Source reference       | <a href="https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=M&amp;cmdCode=721061&amp;motCode=0&amp;customsCode=C00">https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=M&amp;cmdCode=721061&amp;motCode=0&amp;customsCode=C00</a> |

### P-COMTRADE-721061-PARTNERS-ATTEMPT-2

UN Comtrade Saudi 2024 import partner attempt for HS 721061

| Source field | Value       |
|--------------|-------------|
| Source       | UN Comtrade |
| Class        | B           |

| Source field           | Value                                                                                                                                                                                                                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status                 | unresolved                                                                                                                                                                                                                                                                                                          |
| Source period          | 2024                                                                                                                                                                                                                                                                                                                |
| Retrieval date         | 2026-09-13                                                                                                                                                                                                                                                                                                          |
| Reviewer status        | unconfirmed_by_responsible_authority                                                                                                                                                                                                                                                                                |
| Transformation         | PARTNER_DETAIL_ATTEMPT_SUPERSEDED:COVERAGE_INDETERMINATE; unit un_comtrade/e71395bbc925/20260913T010452Z; http_status 200; normalization_status NORMALIZED; superseded by COMPLETE unit un_comtrade/050a03afdc48/20260913T015620Z. This passport evidences the attempt, not trade.                                  |
| Supporting claims      | <ul style="list-style-type: none"> <li>SUPPLIER_CONCENTRATION</li> </ul>                                                                                                                                                                                                                                            |
| Contradiction register | Unavailable                                                                                                                                                                                                                                                                                                         |
| Source reference       | <a href="https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=M&amp;cmdCode=721061&amp;motCode=0&amp;customsCode=C00">https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=M&amp;cmdCode=721061&amp;motCode=0&amp;customsCode=C00</a> |

## P-COMTRADE-721061-X

UN Comtrade Saudi exports rows for HS 721061

| Source field           | Value                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Source                 | UN Comtrade                                                                                                              |
| Class                  | B                                                                                                                        |
| Status                 | calculated                                                                                                               |
| Source period          | 2022/2023/2024                                                                                                           |
| Retrieval date         | 2026-09-12                                                                                                               |
| Reviewer status        | unconfirmed_by_responsible_authority                                                                                     |
| Transformation         | LATEST_REVISION_ONLY H6 rows; trade value divided by 1,000,000 to USD million and net weight divided by 1,000,000 to kt. |
| Supporting claims      | <ul style="list-style-type: none"> <li>EXPORT_IMPORT_RATIO</li> <li>TRADE_QUANTITY</li> <li>TRADE_VALUE</li> </ul>       |
| Contradiction register | Unavailable                                                                                                              |

| Source field     | Value                                                                                                                                                                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source reference | <a href="https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=X&amp;cmdCode=AG6&amp;partnerCode=0&amp;motCode=0&amp;customsCode=C00">https://comtradeapi.un.org/data/v1/get/C/A/HS?reporterCode=682&amp;period=2024&amp;flowCode=X&amp;cmdCode=AG6&amp;partnerCode=0&amp;motCode=0&amp;customsCode=C00</a> |

### P-WCO-721061

HS Nomenclature 2022 — Chapter 72

| Source field           | Value                                                                                                                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source                 | wco_hs_nomenclature                                                                                                                                                                                                                                                                                       |
| Class                  | B                                                                                                                                                                                                                                                                                                         |
| Status                 | observed                                                                                                                                                                                                                                                                                                  |
| Source period          | 2022                                                                                                                                                                                                                                                                                                      |
| Retrieval date         | 2026-09-12                                                                                                                                                                                                                                                                                                |
| Reviewer status        | unconfirmed_by_responsible_authority                                                                                                                                                                                                                                                                      |
| Transformation         | Verbatim span from DocumentRecord DOC-WCO-HS-NOMENCLATURE-5530bdf97060-8da42b5156a2.                                                                                                                                                                                                                      |
| Supporting claims      | <ul style="list-style-type: none"> <li>TARGET_PRODUCT_IDENTITY</li> </ul>                                                                                                                                                                                                                                 |
| Contradiction register | Unavailable                                                                                                                                                                                                                                                                                               |
| Source reference       | <a href="https://www.wcoomd.org/-/media/wco/public/global/pdf/topics/nomenclature/instruments-and-tools/hs-nomenclature-2022/2022/1572_2022e.pdf?la=en">https://www.wcoomd.org/-/media/wco/public/global/pdf/topics/nomenclature/instruments-and-tools/hs-nomenclature-2022/2022/1572_2022e.pdf?la=en</a> |

### P-WITS-721061-PARTNERS-ATTEMPT

World Bank WITS Saudi 2024 import partner attempt for HS 721061

| Source field    | Value                                |
|-----------------|--------------------------------------|
| Source          | World Bank WITS                      |
| Class           | B                                    |
| Status          | unresolved                           |
| Source period   | 2024                                 |
| Retrieval date  | 2026-09-12                           |
| Reviewer status | unconfirmed_by_responsible_authority |

| Source field           | Value                                                                                                                                                                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transformation         | PARTNER_DETAIL_ATTEMPT_SUPERSEDED:FORMAT_NOT_PARSEABLE; unit wits_trade/ed5759a33b4f/20260912T233403Z; http_status 200; normalization_status UNPARSED; superseded by COMPLETE unit un_comtrade/050a03afdc48/20260913T015620Z. This passport evidences the attempt, not trade.                     |
| Supporting claims      | <ul style="list-style-type: none"><li>SUPPLIER_CONCENTRATION</li></ul>                                                                                                                                                                                                                            |
| Contradiction register | Unavailable                                                                                                                                                                                                                                                                                       |
| Source reference       | <a href="https://wits.worldbank.org/Error.aspx?aspxerrorpath=/trade/comtrade/en/country/SAU/year/2024/tradeflow/Imports/partner/ALL/product/721061">https://wits.worldbank.org/Error.aspx?aspxerrorpath=/trade/comtrade/en/country/SAU/year/2024/tradeflow/Imports/partner/ALL/product/721061</a> |

SYN-MINISTRY-GALVALUME-001::capability\_states

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Capability States

| Source field           | Value                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------|
| Source                 | DEMO_GENERATOR                                                                             |
| Class                  | D                                                                                          |
| Status                 | synthetic                                                                                  |
| Source period          | Unavailable                                                                                |
| Retrieval date         | Unavailable                                                                                |
| Reviewer status        | Unavailable                                                                                |
| Transformation         | Unavailable                                                                                |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: capability_states</li></ul> |
| Contradiction register | Unavailable                                                                                |

SYN-MINISTRY-GALVALUME-001::class\_if\_confirmed

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Class If Confirmed

| Source field           | Value                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------|
| Source                 | DEMO_GENERATOR                                                                              |
| Class                  | D                                                                                           |
| Status                 | synthetic                                                                                   |
| Source period          | Unavailable                                                                                 |
| Retrieval date         | Unavailable                                                                                 |
| Reviewer status        | Unavailable                                                                                 |
| Transformation         | Unavailable                                                                                 |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: class_if_confirmed</li></ul> |
| Contradiction register | Unavailable                                                                                 |

SYN-MINISTRY-GALVALUME-001::counterfactual

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Counterfactual

| Source field    | Value          |
|-----------------|----------------|
| Source          | DEMO_GENERATOR |
| Class           | D              |
| Status          | synthetic      |
| Source period   | Unavailable    |
| Retrieval date  | Unavailable    |
| Reviewer status | Unavailable    |
| Transformation  | Unavailable    |

| Source field           | Value                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------|
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: counterfactual</li></ul> |
| Contradiction register | Unavailable                                                                             |

**SYN-MINISTRY-GALVOLUME-001::decision\_specific\_hard\_gates**

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

**SIMULATED — NOT MINISTRY EVIDENCE**

Decision Specific Hard Gates

| Source field           | Value                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------|
| Source                 | DEMO_GENERATOR                                                                                        |
| Class                  | D                                                                                                     |
| Status                 | synthetic                                                                                             |
| Source period          | Unavailable                                                                                           |
| Retrieval date         | Unavailable                                                                                           |
| Reviewer status        | Unavailable                                                                                           |
| Transformation         | Unavailable                                                                                           |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: decision_specific_hard_gates</li></ul> |
| Contradiction register | Unavailable                                                                                           |

**SYN-MINISTRY-GALVOLUME-001::demand**

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

**SIMULATED — NOT MINISTRY EVIDENCE**

Demand

| Source field  | Value          |
|---------------|----------------|
| Source        | DEMO_GENERATOR |
| Class         | D              |
| Status        | synthetic      |
| Source period | Unavailable    |

| Source field           | Value                                                                             |
|------------------------|-----------------------------------------------------------------------------------|
| Retrieval date         | Unavailable                                                                       |
| Reviewer status        | Unavailable                                                                       |
| Transformation         | Unavailable                                                                       |
| Supporting claims      | <ul style="list-style-type: none"> <li>Simulation branch input: demand</li> </ul> |
| Contradiction register | Unavailable                                                                       |

**SYN-MINISTRY-GALVOLUME-001::economics**

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

**SIMULATED — NOT MINISTRY EVIDENCE**

Economics

| Source field           | Value                                                                                |
|------------------------|--------------------------------------------------------------------------------------|
| Source                 | DEMO_GENERATOR                                                                       |
| Class                  | D                                                                                    |
| Status                 | synthetic                                                                            |
| Source period          | Unavailable                                                                          |
| Retrieval date         | Unavailable                                                                          |
| Reviewer status        | Unavailable                                                                          |
| Transformation         | Unavailable                                                                          |
| Supporting claims      | <ul style="list-style-type: none"> <li>Simulation branch input: economics</li> </ul> |
| Contradiction register | Unavailable                                                                          |

**SYN-MINISTRY-GALVOLUME-001::equivalence**

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

**SIMULATED — NOT MINISTRY EVIDENCE**

Equivalence

| Source field | Value          |
|--------------|----------------|
| Source       | DEMO_GENERATOR |

| Source field           | Value                                                                                |
|------------------------|--------------------------------------------------------------------------------------|
| Class                  | D                                                                                    |
| Status                 | synthetic                                                                            |
| Source period          | Unavailable                                                                          |
| Retrieval date         | Unavailable                                                                          |
| Reviewer status        | Unavailable                                                                          |
| Transformation         | Unavailable                                                                          |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: equivalence</li></ul> |
| Contradiction register | Unavailable                                                                          |

SYN-MINISTRY-GALVALUME-001::evsi

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Evsi

| Source field           | Value                                                                         |
|------------------------|-------------------------------------------------------------------------------|
| Source                 | DEMO_GENERATOR                                                                |
| Class                  | D                                                                             |
| Status                 | synthetic                                                                     |
| Source period          | Unavailable                                                                   |
| Retrieval date         | Unavailable                                                                   |
| Reviewer status        | Unavailable                                                                   |
| Transformation         | Unavailable                                                                   |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: evsi</li></ul> |
| Contradiction register | Unavailable                                                                   |

SYN-MINISTRY-GALVOLUME-001::hard\_exclusion\_inputs

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Exclusion Inputs

| Source field           | Value                                                                                          |
|------------------------|------------------------------------------------------------------------------------------------|
| Source                 | DEMO_GENERATOR                                                                                 |
| Class                  | D                                                                                              |
| Status                 | synthetic                                                                                      |
| Source period          | Unavailable                                                                                    |
| Retrieval date         | Unavailable                                                                                    |
| Reviewer status        | Unavailable                                                                                    |
| Transformation         | Unavailable                                                                                    |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: hard_exclusion_inputs</li></ul> |
| Contradiction register | Unavailable                                                                                    |

SYN-MINISTRY-GALVOLUME-001::hard\_gates

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Gates

| Source field    | Value          |
|-----------------|----------------|
| Source          | DEMO_GENERATOR |
| Class           | D              |
| Status          | synthetic      |
| Source period   | Unavailable    |
| Retrieval date  | Unavailable    |
| Reviewer status | Unavailable    |
| Transformation  | Unavailable    |

| Source field           | Value                                                                               |
|------------------------|-------------------------------------------------------------------------------------|
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: hard_gates</li></ul> |
| Contradiction register | Unavailable                                                                         |

**SYN-MINISTRY-GALVOLUME-001::plant\_line**

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

**SIMULATED — NOT MINISTRY EVIDENCE**

Plant Line

| Source field           | Value                                                                               |
|------------------------|-------------------------------------------------------------------------------------|
| Source                 | DEMO_GENERATOR                                                                      |
| Class                  | D                                                                                   |
| Status                 | synthetic                                                                           |
| Source period          | Unavailable                                                                         |
| Retrieval date         | Unavailable                                                                         |
| Reviewer status        | Unavailable                                                                         |
| Transformation         | Unavailable                                                                         |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: plant_line</li></ul> |
| Contradiction register | Unavailable                                                                         |

**SYN-MINISTRY-GALVOLUME-001::route\_evidence**

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

**SIMULATED — NOT MINISTRY EVIDENCE**

Route Evidence

| Source field  | Value          |
|---------------|----------------|
| Source        | DEMO_GENERATOR |
| Class         | D              |
| Status        | synthetic      |
| Source period | Unavailable    |

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة  
**SIMULATED — NOT MINISTRY EVIDENCE**

| Source field           | Value                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------|
| Retrieval date         | Unavailable                                                                             |
| Reviewer status        | Unavailable                                                                             |
| Transformation         | Unavailable                                                                             |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: route_evidence</li></ul> |
| Contradiction register | Unavailable                                                                             |

**SYN-MINISTRY-GALVOLUME-001::target\_specification**

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

**SIMULATED — NOT MINISTRY EVIDENCE**

Target Specification

| Source field           | Value                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------|
| Source                 | DEMO_GENERATOR                                                                                |
| Class                  | D                                                                                             |
| Status                 | synthetic                                                                                     |
| Source period          | Unavailable                                                                                   |
| Retrieval date         | Unavailable                                                                                   |
| Reviewer status        | Unavailable                                                                                   |
| Transformation         | Unavailable                                                                                   |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: target_specification</li></ul> |
| Contradiction register | Unavailable                                                                                   |

**SYN-MINISTRY-GALVOLUME-001::upgrade**

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

**SIMULATED — NOT MINISTRY EVIDENCE**

Upgrade

| Source field | Value          |
|--------------|----------------|
| Source       | DEMO_GENERATOR |

| Source field           | Value                                                                            |
|------------------------|----------------------------------------------------------------------------------|
| Class                  | D                                                                                |
| Status                 | synthetic                                                                        |
| Source period          | Unavailable                                                                      |
| Retrieval date         | Unavailable                                                                      |
| Reviewer status        | Unavailable                                                                      |
| Transformation         | Unavailable                                                                      |
| Supporting claims      | <ul style="list-style-type: none"><li>Simulation branch input: upgrade</li></ul> |
| Contradiction register | Unavailable                                                                      |

## Conditions, kill conditions and next facts

Class D simulation · SYN-MINISTRY-GALVALUME-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

**SIMULATED — NOT MINISTRY EVIDENCE**

### Decision conditions

- Achieve third-party certification to the target standard and first customer acceptance within 12 months.
- Qualification cost-share is milestone-based and sunset-bound.

### Kill conditions

- Committed target-specification demand falls below the minimum efficient qualification volume.
- Certification is not achieved by the contractual milestone.

### Next evidence actions

- None

| Source field             | Value | Unit  |
|--------------------------|-------|-------|
| Approximate EVSI         | 44.2  | SAR m |
| Route-change probability | 0.6   |       |
| Value difference (SAR m) | 82    |       |

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة  
**SIMULATED — NOT MINISTRY EVIDENCE**

|                                   |                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------|
| Evidence acquisition cost (SAR m) | 2                                                                                    |
| Delay cost (SAR m)                | 3                                                                                    |
| Approximate EVSI                  | 44.2                                                                                 |
| Positive result                   | Yes                                                                                  |
| Next fact to establish            | Confirm the incumbent's coating-mass capability and buyer qualification requirements |

## Decision history and expert overrides

None recorded; no completed review is implied.

## Authority and reproducibility

### Authority and versions

| Source reference | Value                                                                                                                                           |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Methodology      | docs/authority/Industrial_Opportunity_Resolution_Methodology_Final_KSA.docx<br>5717cbd42acc9947ce5e450013719275acb7ed1470847b21fb2cc547c8ac4ce9 |
| Project          | 0.2.0                                                                                                                                           |
| Thresholds       | 1.3.0                                                                                                                                           |
| Sector profiles  | 1.1.0                                                                                                                                           |
| Evidence policy  | 1.4.0                                                                                                                                           |
| UI strings       | 1.7.0                                                                                                                                           |

### Evidence boundary

|                                           |      |
|-------------------------------------------|------|
| Public decision uses public evidence only | Yes  |
| Synthetic branch isolation                | Yes  |
| Scenario reconciliation                   |      |
| Status                                    | PASS |

|                                          |                                                                        |
|------------------------------------------|------------------------------------------------------------------------|
| Scenario identifier                      | SYN-MINISTRY-GALVALUME-001                                             |
| Opportunity identifier                   | SAU-H6-721061                                                          |
| Public snapshot identifier               | PUBLIC-SAU-H6-721061-2026-09-12                                        |
| Recorded checks                          |                                                                        |
| • Rule identifier                        | target_spec_demand_within_public_imports                               |
| Source                                   | Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195                    |
| Recorded formula                         | target_spec_demand_kt <= latest_public_imports_kt                      |
| Recorded result                          | PASS                                                                   |
| Blocking check                           | Yes                                                                    |
| Source inputs                            |                                                                        |
| Target-specification demand              | 48                                                                     |
| Latest public observation year           | 2024                                                                   |
| Latest public imports (kt)               | 93.232034                                                              |
| Source explanation                       | 48 kt <= 93.232 kt.                                                    |
| • Rule identifier                        | line_nameplate_within_disclosed_public_capacity                        |
| Source                                   | Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195                    |
| Recorded formula                         | plant_line.nameplate_kt <= sum(non_null installed_capacity_tpy) / 1000 |
| Recorded result                          | Not applicable                                                         |
| Blocking check                           | Yes                                                                    |
| Source inputs                            |                                                                        |
| Line nominal capacity (kt)               | 120                                                                    |
| Disclosed installed capacity (t/year)    | ◦ None recorded; no completed review is implied.                       |
| Publicly disclosed nominal capacity (kt) | Unavailable                                                            |
| Expansion assumption applied             | No                                                                     |
| Source explanation                       | No numeric public installed nameplate is disclosed.                    |

|                                |                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| • <b>Rule identifier</b>       | capacity_factors_within_unit_interval                                                                                                                                   |
| <b>Source</b>                  | Core 06 §10 item 3 lines 226-233; Core 06 §6.2 lines 148-156; Core 05 §8 lines 197-203                                                                                  |
| <b>Recorded formula</b>        | for every capacity factor: $0 \leq \text{factor} \leq 1$                                                                                                                |
| <b>Recorded result</b>         | PASS                                                                                                                                                                    |
| <b>Blocking check</b>          | Yes                                                                                                                                                                     |
| <b>Source inputs</b>           |                                                                                                                                                                         |
| <b>Availability factor</b>     | 0.9                                                                                                                                                                     |
| <b>Yield factor</b>            | 0.94                                                                                                                                                                    |
| <b>Qualification share</b>     | 0.35                                                                                                                                                                    |
| <b>Market allocation share</b> | 0.7                                                                                                                                                                     |
| <b>Current utilisation</b>     | 0.86                                                                                                                                                                    |
| <b>Source explanation</b>      | All declared capacity factors are within [0,1].                                                                                                                         |
| • <b>Rule identifier</b>       | qualified_availability_within_physical_output                                                                                                                           |
| <b>Source</b>                  | Core 06 §5.1 lines 92-101; Core 06 §10 item 3 lines 226-233                                                                                                             |
| <b>Recorded formula</b>        | $\text{qualified\_available\_kt} \leq \text{nameplate\_kt} * \text{availability} * \text{yield}$                                                                        |
| <b>Recorded result</b>         | Not applicable                                                                                                                                                          |
| <b>Blocking check</b>          | Yes                                                                                                                                                                     |
| <b>Source inputs</b>           |                                                                                                                                                                         |
| <b>Source explanation</b>      | No qualified_available_kt is declared.                                                                                                                                  |
| • <b>Rule identifier</b>       | demand_layers_remain_separate                                                                                                                                           |
| <b>Source</b>                  | Core 04 §2.6 lines 102-112                                                                                                                                              |
| <b>Recorded formula</b>        | observe $\text{downside\_demand\_kt} \leq \text{target\_spec\_demand\_kt}$ and $\text{committed\_demand\_kt} \leq \text{target\_spec\_demand\_kt}$ ; no decision effect |
| <b>Recorded result</b>         | INFORMATIONAL                                                                                                                                                           |
| <b>Blocking check</b>          | No                                                                                                                                                                      |

|                                       |                                                                            |
|---------------------------------------|----------------------------------------------------------------------------|
| <b>Source inputs</b>                  |                                                                            |
| Target-specification demand           | 48                                                                         |
| Downside demand                       | 46                                                                         |
| Committed demand                      | 30                                                                         |
| Base demand                           | 93                                                                         |
| Downside demand within target demand  | Yes                                                                        |
| Committed demand within target demand | Yes                                                                        |
| Source explanation                    | Demand layers are reported separately; no ordering constraint is enforced. |
| • Rule identifier                     | tariff_line_allocation_sums_to_public_hs6_total                            |
| Source                                | Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195                        |
| Recorded formula                      | sum(tariff_line_allocation.lines.quantity_kt) == latest public imports_kt  |
| Recorded result                       | Not applicable                                                             |
| Blocking check                        | Yes                                                                        |
| <b>Source inputs</b>                  |                                                                            |
| Source explanation                    | No tariff_line_allocation block is declared.                               |
| • Rule identifier                     | buyer_allocation_within_public_imports                                     |
| Source                                | Core 06 §5.1 lines 92-101                                                  |
| Recorded formula                      | sum(buyer_allocation.buyers.quantity_kt) <= latest public imports_kt       |
| Recorded result                       | Not applicable                                                             |
| Blocking check                        | Yes                                                                        |
| <b>Source inputs</b>                  |                                                                            |
| Source explanation                    | No buyer_allocation block is declared.                                     |
| • Rule identifier                     | expansion_assumption_disclosed_and_bounded                                 |

|                                     |                                                                                                                                                    |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Source</b>                       | Core 06 §5.1 lines 92-101; KL-38                                                                                                                   |
| <b>Recorded formula</b>             | planned_nameplate_kt >= disclosed public nameplate;<br>commissioning_year >= public as_of year; plant_line.nameplate_kt<br><= planned_nameplate_kt |
| <b>Recorded result</b>              | Not applicable                                                                                                                                     |
| <b>Blocking check</b>               | Yes                                                                                                                                                |
| <b>Source inputs</b>                |                                                                                                                                                    |
| <b>Source explanation</b>           | No expansion_assumption block is declared.                                                                                                         |
| • <b>Rule identifier</b>            | retained_flows_reconcile_to_public_trade                                                                                                           |
| <b>Source</b>                       | Core 06 §5.1; methodology §3.3                                                                                                                     |
| <b>Recorded formula</b>             | retained_imports_kt <= imports_kt; retained + reexports ==<br>imports when reexports numeric; domestic_origin_exports_kt <=<br>public exports_kt   |
| <b>Recorded result</b>              | Not applicable                                                                                                                                     |
| <b>Blocking check</b>               | Yes                                                                                                                                                |
| <b>Source inputs</b>                |                                                                                                                                                    |
| <b>Source explanation</b>           | No production_and_retained_flows block is declared.                                                                                                |
| • <b>Rule identifier</b>            | base_demand_and_commitment_probability_valid                                                                                                       |
| <b>Source</b>                       | Core 06 §5.2; Core 07 §3 R8                                                                                                                        |
| <b>Recorded formula</b>             | 0 <= commitment_probability <= 1; base_demand_kt > 0;<br>minimum_efficient_scale_kt > 0 when declared                                              |
| <b>Recorded result</b>              | PASS                                                                                                                                               |
| <b>Blocking check</b>               | Yes                                                                                                                                                |
| <b>Source inputs</b>                |                                                                                                                                                    |
| <b>Base demand</b>                  | 93                                                                                                                                                 |
| <b>Commitment probability</b>       | 0.8                                                                                                                                                |
| <b>Minimum efficient scale (kt)</b> | 30                                                                                                                                                 |
| <b>Source explanation</b>           | Base demand, commitment probability and MES inputs are valid.                                                                                      |

|                                                   |                                                           |
|---------------------------------------------------|-----------------------------------------------------------|
| • <b>Rule identifier</b>                          | shared_enabler_declaration_valid                          |
| <b>Source</b>                                     | Core 06 scenario contract 2.1.0 and methodology §8.3      |
| <b>Recorded formula</b>                           | explicit route valuation and complete Class-D declaration |
| <b>Recorded result</b>                            | Not applicable                                            |
| <b>Blocking check</b>                             | Yes                                                       |
| <b>Source inputs</b>                              |                                                           |
| <b>Source explanation</b>                         | No shared-enabler declaration is present.                 |
| <b>Latest imports (USD million)</b>               | 71.149266                                                 |
| <b>Latest imports (kt)</b>                        | 93.232034                                                 |
| <b>Public decision unchanged after simulation</b> | Yes                                                       |
| <b>Scenario expected-result comparison</b>        |                                                           |
| <b>Expected result</b>                            |                                                           |
| <b>State</b>                                      | ADVANCE                                                   |
| <b>Route code</b>                                 | 3                                                         |
| <b>Actual result</b>                              |                                                           |
| <b>State</b>                                      | ADVANCE                                                   |
| <b>Route code</b>                                 | 3                                                         |
| <b>Results match</b>                              | Yes                                                       |